

DIFFUSION OF MANGANESE IN α -IRON SINGLE CRYSTALS OF DIFFERENT PURITY

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Abstract—Diffusion coefficients of manganese into pure and alloyed α -iron single crystals were determined in the temperature range 778.3–901°C.

The Arrhenius curves for diffusion into pure α -iron and α -iron alloyed with Ti or V are linear and yield the same activation energy $Q = 60.1$ kcal/mole ± 1.5 per cent. For diffusion into an α -iron alloyed with copper $\ln D = f(1/T)$ is nonlinear. A linear dependence between the preexponential factors and the concentrations of oxygen and carbon dissolved on interstitial sites was found.

The results are discussed using the concept of free and associated vacancies. A compilation of data for the diffusion of metals in α -iron is given; it does not conform to any of the existing theories.

1. INTRODUCTION

DURING a research program about the influence of foreign atoms on self-diffusion in high purity α -iron solute diffusion of ^{54}Mn in α -iron single crystals was also investigated.

Regarding volume hetero-diffusion in a matrix element two questions have to be discussed:

1. How is solute diffusion affected by the presence of other elements in the matrix?
2. In which way does the solute diffusion coefficient vary in a certain matrix if the diffusing element is changed?

No theories and only few experimental contributions have been given on the influence of additional foreign atoms on volume hetero diffusion. So far only experimental data of Sonder[1] and experimental results together with their qualitative explanation by Helfmeier and Feller-Kniepmeier[2] are available. Several theories are given for the mechanism of hetero diffusion via vacancies. Two different types of approaches can be distinguished: One is to apply lattice kinetic theory to the problem of solute diffusion. By doing this Lidiard[3] calculated the diffusion coefficient as a function of jump frequencies. This approach, however, does not give any evidence on how the jump frequencies depend on the nature of the diffusing element. The

second method is more empirical. Here it is attempted to connect the different diffusion coefficients of elements in a certain matrix with their physical properties. Swalin[4] postulated that the activation energy of diffusion depends on the ratio of the atomic radii for solute and matrix atoms. In the model of Lazarus[5] the changes in activation energy for different solute atoms are due to their electric charge in the matrix element. This theory has been extended[6–8] and was applied to homovalent elements by Le Claire [9]. Up to now the only values for diffusion of manganese in α -iron are given by Nohara and Hirano[10]. Using polycrystalline iron of 99.97 per cent purity and applying the method of Gruzin[11] these authors find an activation energy of 52.5 ± 2.3 kcal/mole and a pre-exponential factor of 0.35 cm²/sec.

2. EXPERIMENTAL

For our experiments raw material of different purity was used: rods (12 cm long, ϕ 5 mm) of 99.99% Fe in the metallic compounds and a powder of 99.999% iron in the metallic compounds. Table 1 gives the analysis of the different iron raw materials. The further purification of the iron as well as the preparation of single crystals both of pure α -iron and of α -iron alloyed with 0.09 at.% vanadium,

Table 1

Analysis of Fe powder, initial state						
	Si	Mn	Cu	Mg	Co	
In weight ppm	2	5	3	1	1	

Analysis of Fe rods, initial state ($R_{293^{\circ}\text{K}}/R_{4.2^{\circ}\text{K}} = 40$)									
	Si	Mn	Cu	Ag	Mg	Al	C	N	O
In weight ppm	3	2	5	1	2	1	30	20	10

0.11 at.% titanium, and 0.27 at.% copper respectively is published in detail in other papers [12, 14]. Table 2 shows a compilation of the specimens used. For the pure iron crystals the resistivity ratio $R_{293^\circ\text{K}}/R_{4.2^\circ\text{K}}$ is given as a purity criterion. Alloyed single crystals were prepared from iron with a resistivity ratio of 500. The concentration of oxygen and carbon is given in a relative scale. An activation analysis carried out for Fe III gave a carbon content of 1 ± 0.7 ppm and an oxygen content of 7 ± 5 ppm.

^{54}Mn was generated by neutron bombardment ($10d$, 10^{13} n/sec) of a pure iron wire. The main isotopes produced were:

^{59}Fe ($t_{1/2} = 45d$; β -radiation; γ -radiation: 1.10 MeV, 1.29 MeV)

^{55}Fe ($t_{1/2} = 2.6y$; K -radiation: 5.9 keV)

^{54}Mn ($t_{1/2} = 313d$; γ -radiation: 0.84 MeV).

At a vacuum of 10^{-7} torr the activated wire was evaporated on to the front surface of the cylindrical single crystals (length 5 mm, ϕ 5 mm), the thickness being about $0.1\mu\text{m}$. At a vacuum of $5 \cdot 10^{-8}$ torr the specimens were

sealed in quartz capsules. The temperature of the annealing furnace could be controlled to $\pm 0.2^\circ\text{C}$. Temperature adjustability was also $\pm 0.2^\circ\text{C}$. Taking into account aging of the PtRh/Pt thermocouple and drifts in the electronic device, the temperature constancy was 0.5°C over the length of the diffusion times. Diffusion runs were made between 778.3°C and 901°C . After diffusion the cylindrical surfaces of the crystals were turned off by $50\mu\text{m}$. Sections of about $1\mu\text{m}$ thickness were cut by a microtome, the exact thickness of the chips being determined by their weights. The activity of Mn 54 in the chips was counted using a NaI(Tl) scintillation counter and a discriminator unit. The resolution of the crystal was good enough to clearly separate the Mn and Fe peaks. For every chip at least 10^4 impulses were counted.

3. RESULTS

For the experimental conditions applied the solution of the diffusion equation is

$$\ln I = -x^2/(4Dt) + K \quad (1)$$

Table 2. Concentration of impurities (C and O relative values) and fraction of residual resistance $R_{293^\circ\text{K}}/R_{4.2^\circ\text{K}}$

Material	($R_{293^\circ\text{K}}/R_{4.2^\circ\text{K}}$)	C [Relative]	O [Relative]	V [at.%]	Ti [at.%]	Cu [at.%]
Fe I	150	15	8			
Fe II	500	1	4			
Fe III	1500	1	1			
Fe-V		2.5	3	0.09	—	—
Fe-Ti		1.5	2	—	0.11	—
Fe-Cu		1	1.5	—	—	0.27

I is the specific ^{54}Mn activity in a thin section the center of which is x centimeters from the surface, D the diffusion coefficient, t the diffusion time and K a constant. A plot of $\ln I_{\text{Mn}}$ vs. x^2 can therefore be used to determine the diffusion coefficient. Figure 1 shows this type of plot for the three iron charges of different purity. At identical diffusion time and temperature the diffusion coefficient of manganese is smallest in the iron of highest purity and increases with decreasing purity of the matrix. The same dependence is found for all the diffusion runs within the temperature range used. Table 3 gives a compilation of the diffusion coefficients obtained. For Fe II the

plot $\ln D = f(1/T)$ is shown in Fig. 2. Diffusion data at five temperatures between 778.3 and 901°C were obtained using Fe II as matrix element. At 839.7°C two specimens were annealed with different diffusion times. Their values differ by ± 2.5 per cent. The diffusion coefficients at the four higher temperatures were used to calculate the least square line. The results are:

$$Q_{\text{II}} = 60.1 \text{ kcal/mole}$$

$$D_{0\text{II}} = 0.78 \text{ cm}^2/\text{sec.}$$

Diffusion coefficients between 809 and 901°C fit this line within ± 3 per cent while the value measured at 778.3°C deviates from the extrapolated line by -20 per cent.

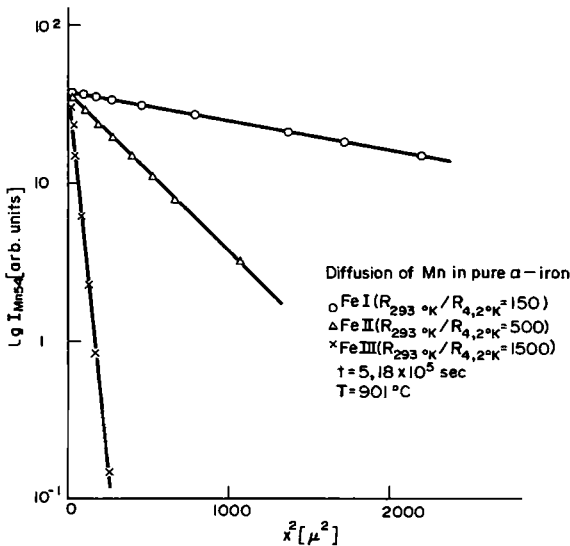


Fig. 1. Plot of $\ln I$ vs. x^2 for diffusion of Mn in α -iron.

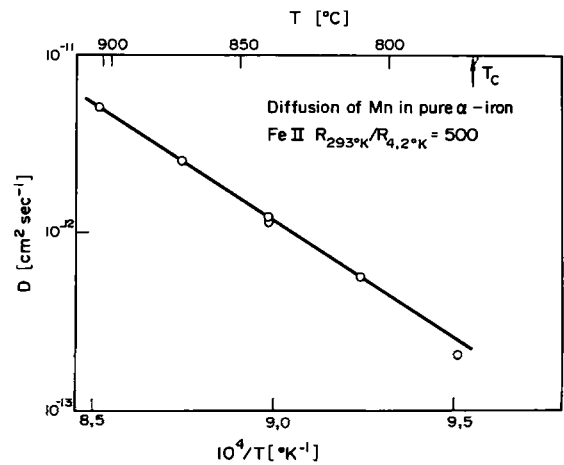


Fig. 2. Solute diffusion coefficients of Mn into pure α -iron (Fe II) as a function of $1/T$.

Table 3. Results for solute diffusion of Mn in α -iron

Material		I	II	III	V	Ti	Cu
T [°C]							
D [cm ² /sec]	901	232	50.4	6.75	28.2	39.2	34.2
	870.3	—	25.5	—	—	—	—
	855	—	—	—	9.77	1.39	—
	10 ⁻¹³	839.7	56.6	12.3	—	—	—
			11.8	1.67			
Q [kcal/mole] D_0 [cm ² /sec]	809	—	5.70	—	3.11	4.43	3.76
	778.3	9.5	2.07	0.28	1.15	1.63	1.74
			59.6	59.0	60.5	60.1	—
		3.6	0.78	0.11	0.43	0.60	—

For iron charge I and III as matrix elements diffusion runs were carried out at three different temperatures only (Fig. 3): 778.3, 839 and 901°C. It showed that the data obtained varied from those in Fe II by a constant factor. Diffusion coefficients of Mn in Fe I were always 4.80 ± 3 per cent above those in Fe II while those in Fe III were always 7.44 ± 2 per cent below those in Fe II. This means that the activation energies are the same within experimental error. We therefore postulate $Q_I = Q_{II} = Q_{III} = 60.1$ kcal/mole. The values for D_0 were then calculated using the diffusion coefficients measured at 901°C

$$D_{0I} = 3.6 \text{ cm}^2/\text{sec}$$

$$D_{0III} = 0.11 \text{ cm}^2/\text{sec.}$$

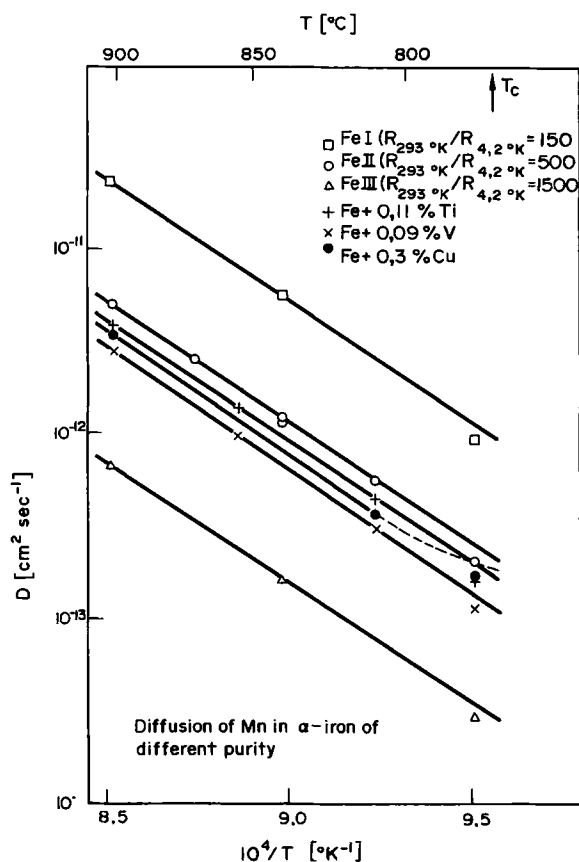


Fig. 3. Solute diffusion coefficient of Mn into pure and alloyed α -iron as a function of $1/T$.

With Fe I and Fe III as matrix elements too the diffusion coefficients obtained at 778.3°C differ by -19 per cent (Fe I) and -22 per cent (Fe III) from the Arrhenius line.

For the diffusion of manganese in α -iron alloyed with titanium and vanadium respectively diffusion runs at four temperatures were carried out (Fig. 3). Taking the data for the three highest temperatures to calculate the least square line an activation energy of $Q_V = 60.5$ kcal/mole for the matrix alloyed with vanadium and of $Q_{Ti} = 60.1$ kcal/mole for the matrix alloyed with titanium was achieved. Within the experimental error these activation energies too coincide with that calculated for the diffusion in Fe II. Taking $Q_{Ti} = Q_V = Q_{II}$ the preexponential factors were calculated to be

$$D_{0Ti} = 0.53 \text{ cm}^2/\text{sec}$$

$$D_{0V} = 0.43 \text{ cm}^2/\text{sec.}$$

The deviation of the diffusion coefficients at 778.3°C from the Arrhenius line obtained for the higher temperature is -24 per cent for the titanium alloyed matrix and -22 per cent for the vanadium alloyed matrix.

For the diffusion of Mn in α -iron alloyed with copper experiments were carried out at 778.3, 839 and 901°C. The values obtained in this system however do not deviate by a constant factor from those measured for the diffusion of Mn in Fe II. The three points can be connected by a curve in several ways:

1. By drawing a straight line with $Q = 60.1$ kcal/mole through the point at 901°C. The coefficient at 839°C is then 7 per cent above and the value at 778.3°C 7 per cent below that line. This type of curve however does not correspond with our experimental error limit of $(\Delta D/D)_{\text{exp}} = \pm 3$ per cent and with the size of the magnetic anomaly at 778.3°C being -19 to -24 per cent for all the other specimens.

2. All three points are used to draw a straight line. The deviation at 809°C is then +4 per cent and at 778.3°C -5 per cent.

3. The diffusion coefficients obtained at the upper two temperatures can be connected by a straight line, the extrapolation of that curve is then 13 per cent above the point at 778.3°C.

Taking into account the relative error of our experiments ($(\Delta D/D)_{\text{exp}} = \pm 3$ per cent) and the magnetic anomaly at 778.3°C determined with the other specimen we feel that the curve can best be drawn by connecting the two values at the higher temperatures and a point which is 20 per cent above the diffusion coefficient measured at 778.3°C. This curve is nonlinear.

A detailed error estimation for the experiments is given in our paper concerning self-diffusion in α -iron [14]. The same considerations are true for the present work. The calculated relative error in the diffusion coefficients is $(\Delta D/D)_{\text{th}} = \pm 4$ per cent, that in the preexponential factors $\Delta D_0/D = \pm 25$ per cent and that in the activation energies $\Delta Q/Q = \pm 1.5$ per cent.

4. DISCUSSION

Our experimental results can be summarized as follows:

1. The diffusion of Mn in pure α -iron can be described by a linear function $\ln D = f(1/T)$.

2. With the exception of the iron matrix alloyed with copper all systems investigated yield the same activation energy. The frequency factors D_0 however depend on the composition of the matrix metal.

3. The Arrhenius curve for diffusion of Mn in α -iron alloyed with copper is nonlinear.

4. Diffusion coefficients measured at 778.3°C deviate from the Arrhenius curve obtained for higher temperatures by about -21 per cent.

1. Provided a vacancy mechanism is operating and under the assumption of weak interaction between solute and vacancy, Lidiard [3] derived for the diffusion coefficient of a solute in a pure matrix:

$$D_i = 2s^2 \frac{w_2(w_1 + 7/2k_1)}{w_1 + w_2 + 7/2k_1} \left(\frac{k_2}{k_1} \right) c_v \quad (2)$$

where s is the jump distance, c_v the mole fraction of free vacancies in a pure metal, w_1 the frequency of exchange between an associated vacancy and a solvent atom, w_2 the frequency of exchange between an associated vacancy and a solute atom, k_1 the dissociation frequency between associated vacancy and unassociated solvent atom and k_2 the association frequency between unassociated vacancy and associated solvent. From equation (2) a temperature dependence of D_i is expected which is the sum of exponential functions, and $\ln D = f(1/T)$ should be nonlinear. As we do find a linear Arrhenius plot for the diffusion of manganese in the pure α -iron single crystals two possible reasons can be assumed. The first is that the temperature range, as it is limited by the α/γ transition and the magnetic transition, is too narrow to distinguish between different types of curves.

The second argument has been given by Koehler and Kino [15] in assuming that one term in equation (2) is dominating. For a positive binding energy and short range interaction between vacancy and solute they obtained $w_1 = w_2$, $w_1 \ll k_1$ and $k_2 = w_0$ where w_0 is the jump frequency between an unassociated vacancy and unassociated solvent atom. Equation (2) is then simplified to:

$$D_i = s^2 \frac{w_0 w_2}{k_1} c_v = s^2 w_2 \exp [-(g_f + \Delta g_i)/kT]. \quad (3)$$

Here g_f is the formation energy of a free vacancy and $-\Delta g_i$ the binding energy of a solute-vacancy pair.

More emphasis is given to the second argument because, e.g., Sonder *et al.* [1] found a linear Arrhenius plot for the diffusion of antimony in silver between 468° and 942°C and Helfmeier and Feller-Kniepmeier [2] get a linear Arrhenius line for the diffusion of copper into nickel single crystals in the temperature range from 775 to 1050°C.

2. Similar results have been obtained for self-diffusion in α -iron single crystals [14] of

different purity. Though the present system is obviously much more complicated it has tried to account for the results using a similar argument. The total amount of vacancies available for material transport in a dilute alloy can be estimated using the equation of Lomer[16]

$$\frac{c_v(c_i)}{c_v(0)} = 1 - zc_i + zc_i \exp(-\Delta g/kT) \quad (4)$$

with $c_v(0)$ = mole fraction of free vacancies

$c_v(c_i)$ = mole fraction of vacancies associated to solute atoms

c_i = mole fraction of solute atoms

z = coordination number of the lattice.

It is seen that for high temperatures $c_v(c_i)$ is approaching $c_v(0)$ while at low temperatures for positive binding energies $c_v(c_i)$ is greater $c_v(0)$. Therefore one ought to expect a positive deviation from the Arrhenius line at lower temperatures. Indeed this behavior has been found for the diffusion of copper in dilute nickel single crystals[2]. From equation (2) an estimation about the diffusion coefficient $D_i(c_j)$ in a metal containing additional impurities c_j can be derived using the assumptions of Koehler and Kino[15] and the equation of Lomer[16]

$$D_i(c_j) \approx 1 + zc_j \exp\left(-\frac{(g + \Delta g_i)}{kT}\right) + zc_j \exp\left[-\frac{(g + \Delta g_i + \Delta g_j)}{kT}\right] \quad (5)$$

D_i being the diffusion coefficient for solute diffusion in the pure matrix via free vacancies and vacancies associated to manganese atoms. Δg_j is the binding energy between vacancies and impurity atoms in the matrix. Depending on temperature the first or second term of equation (5) is dominating, the transition temperature being determined by the magnitude of Δg_j and the impurity concentration c_j . In the low temperature region therefore a

constant activation energy $-[g + \Delta g_i + \Delta g_j]$ may result while the preexponential factor should depend linearly on the impurity concentration. Indeed the temperature of our experiments is 600°C below the melting point and therefore considered to be low. The strong influence of impurities dissolved on interstitial sites is shown in Table 4 and Fig. 4. Within experimental error and taking into account that the interaction of carbon and oxygen with the matrix is not completely equivalent we find a linear dependence of D_0 with $C_g = C_0 + C_c$ as is expected from equation (5). The alteration of the diffusion coefficients of Mn in the FeTi and FeV alloys is then reduced to their content of carbon and oxygen.

Two mechanisms can be discussed to account for the ineffectiveness of Ti and V on the diffusion of Mn in α -Fe. Either the transition temperature could be below the temperature range investigated or Δg_j could

Table 4. Comparison of D_0 values with relative oxygen and carbon concentration on interstitial sites

Material	c_0 [Relative]	c_c [Relative]	D_0 [cm ² /sec]
Fe III	1	1	0.4
Fe-Cu*	1.5	1	0.53
Fe-Ti	2	1.5	0.6
Fe-V	3	2.5	0.43
Fe II	4	1	0.78
Fe I	8	15	3.6

*Calculated from the D value at 901°C and $Q = 60.1$ kcal/mole.

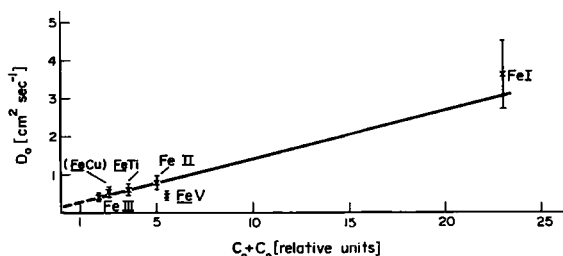


Fig. 4. Preexponential factor D_0 as a function of the sum of the relative oxygen and carbon concentrations.

be equal to zero. We interpret our results by the second explanation. Δg_j being zero can be understood using the argumentation Helfmeier[2] applied to account for his results on the diffusion of copper in dilute nickel single crystals. He found that impurity elements located in the periodic system on the left side of a transition metal had no influence on diffusion coefficients. An explanation was given using Pauling's theory of transition metals.

3. The nonlinearity of the Arrhenius plot for diffusion of manganese in α -iron alloyed with copper is interpreted by assuming that in addition to associated vacancies created by oxygen and carbon dissolved on interstitial sites there are vacancies associated to copper. Again this behavior is accounted for using Helfmeier's[2] explanation. As copper is on the right side of iron in the periodic system it should have a surplus of electrons and therefore a positive binding energy to vacancies in iron.

4. The deviation of the diffusion coefficients measured at 778.3°C from the Arrhenius curves determined for higher temperatures are common for diffusion in α -iron and due to the magnetic transition at the Curie point.

The only data available on diffusion of manganese into α -iron are those of Nohara and Hirano[10]. No explanation can be given for the striking difference between their value for the activation energy and ours.

In a recently published paper Bowen and Leak[17] postulate that for the diffusion of Cr, V and Hf in α -iron the activation energies vary as do the compressibilities of the diffusing elements. They interpret their finding as proof for Swalin's[4] theory as it was taken up by Askill and Gibbs[13]. In Fig. 5 we have plotted the activation energies for diffusion in α -iron as they are given in the literature (Table 5) together with the compressibilities and the atomic radii as a function of the atomic number for the diffusing element. The error limits for the activation energies, as given, are marked. This compilation of the literature

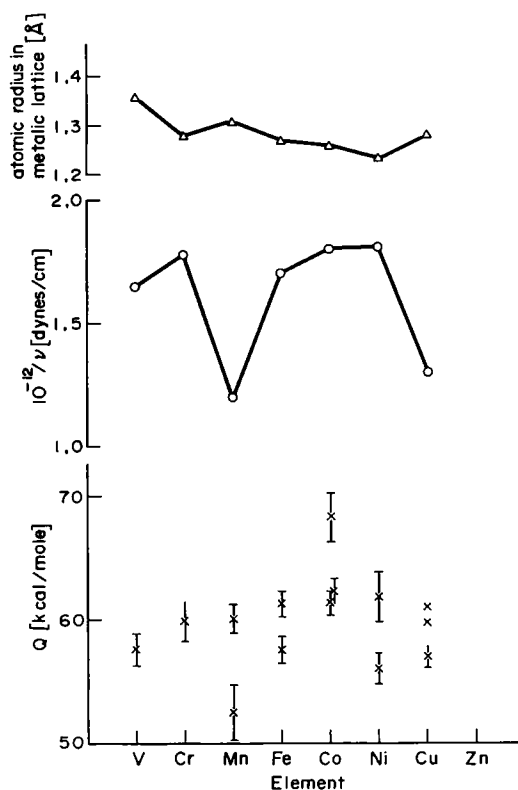


Fig. 5. Activation energy Q of solvent diffusion in α -iron, compressibilities and atomic radii as a function of atomic number (see also Table 5).

Table 5. Compilation of solute diffusion activation energy Q in α -iron from literature (α -iron self diffusion only two representative examples)

Element	Purity of α -iron[%]	Q [kcal/mole]	Ref.
V	99.98	57.6 ± 1.3	[17]
Cr	99.98	59.9 ± 1.6	[17]
	99.97	57.5	[18]
Mn	—	52.5 ± 2.3	[10]
	99.99	60.1	this paper
Fe	99.95	$57.5 \pm 1.8\%$	[19]
	99.99	$61.3 \pm 1.5\%$	[14]
Co	99.95	$61.4 \pm 1.4\%$	[19]
	99.9	62.3 ± 1	[20]
	98.2 + 1.8 V	68.3 ± 2	[21]
Ni	98.2 + 1.8 V	61.9 ± 2	[21]
	99.97	$56 \pm 2\%$	[22]
Cu	99.99	$57.0 \pm 1.5\%$	[23]
	—	59.7	[24] [25]
	99	61	[26]

values does not confirm any of the existing theories. Error limits are generally much too high, and the deviations in the activation energy as determined by different authors for the same element is often worse than 10 per cent. Probably these differences are due to different purities of the metals used for the experiments. The only conclusion one may be able to draw from Fig. 5 is, that the activation energies are relatively high for the diffusion of the transition elements in α -iron and that they go down for the diffusion of copper on one side and for the diffusion of vanadium and chromium on the other side.

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